FIND · BOOK · TRACK · SAVE LIVES

AMBULANCE FINDER AND EMERGENCY RESPONSE

Process drawings for every actor in the system. Each sheet is a swimlane diagram: who does what, in what order, and what happens on every yes, no and timeout. Open "Step reference" under a sheet for the detail behind each box.

DOCUMENT	REVISION	ISSUED	SHEETS
Process flow set	1.0 draft	8 Oct 2026	14

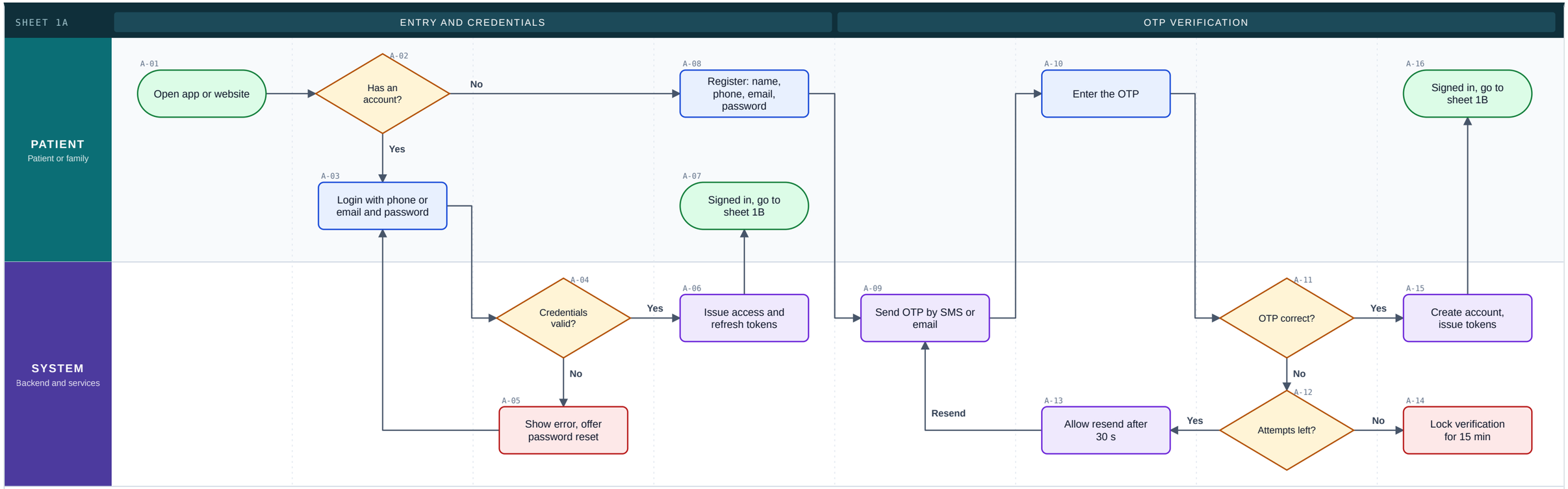
30 s	5 s	5 / 10 / 20 km	3	3
driver response window	GPS update interval	search radius steps	OTP attempts before lock	auto re-dispatch tries

These numbers are assumed defaults used in the drawings. Replace them with your own policy.

PATIENT OR FAMILY

PATIENT SIGN-UP AND SIGN-IN

Returning patients sign in with a password. New patients register and prove their phone or email with an OTP. Both paths end with a signed-in session.



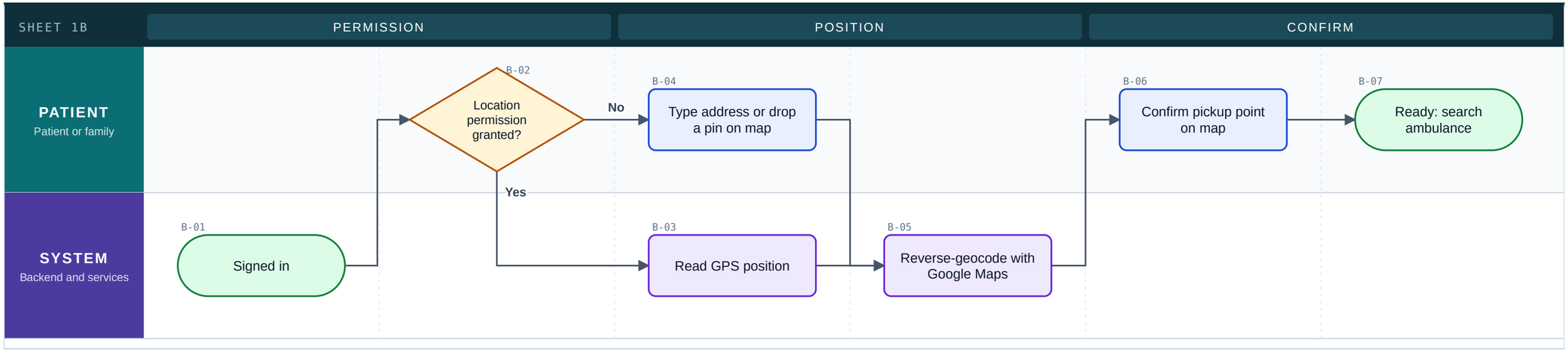
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
A-01	Patient	Open app or website	Web (React) or mobile app.
A-03	Patient	Login with phone or email and password	POST /auth/login
A-04	System	Credentials valid?	Password hash compared; repeated failures lock the account (limit is a policy choice).
A-05	System	Show error, offer password reset	Reset link sent by email or SMS.
A-06	System	Issue access and refresh tokens	JWT access token (short-lived) plus refresh token.
A-08	Patient	Register: name, phone, email, password	POST /auth/register
A-09	System	Send OTP by SMS or email	6-digit code, valid for 5 minutes (assumed).
A-12	System	Attempts left?	Up to 3 attempts (assumed).
A-13	System	Allow resend after 30 s	Resend button is disabled for 30 seconds.
A-14	System	Lock verification for 15 min	Prevents OTP guessing.
A-15	System	Create account, issue tokens	Account becomes active and tokens are issued.

PATIENT OR FAMILY

ALLOW LOCATION AND SET PICKUP POINT

GPS is the fast path. If the patient refuses permission, a typed address or a dropped pin gives the same result.



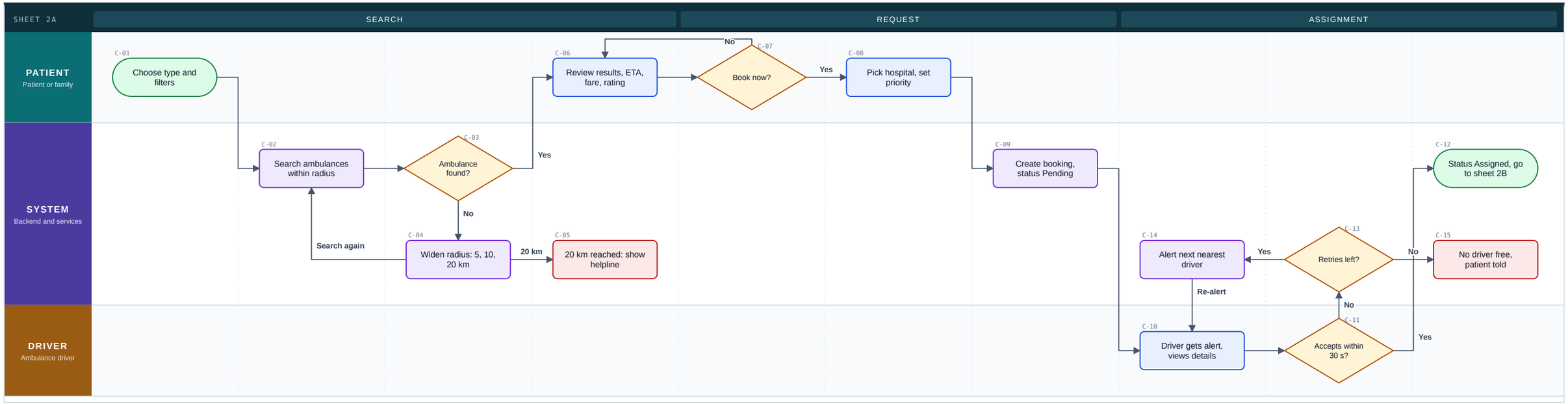
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
B-02	Patient	Location permission granted?	Browser or phone location prompt.
B-03	System	Read GPS position	Accuracy target under 50 m (assumed).
B-05	System	Reverse-geocode with Google Maps	Geocoding API turns coordinates into a readable address.

PATIENT · SYSTEM · DRIVER

BOOKING: SEARCH TO ASSIGNMENT

The patient searches, picks and requests. The system alerts the nearest driver. If nobody accepts within 30 seconds the system moves to the next driver, up to a retry limit.



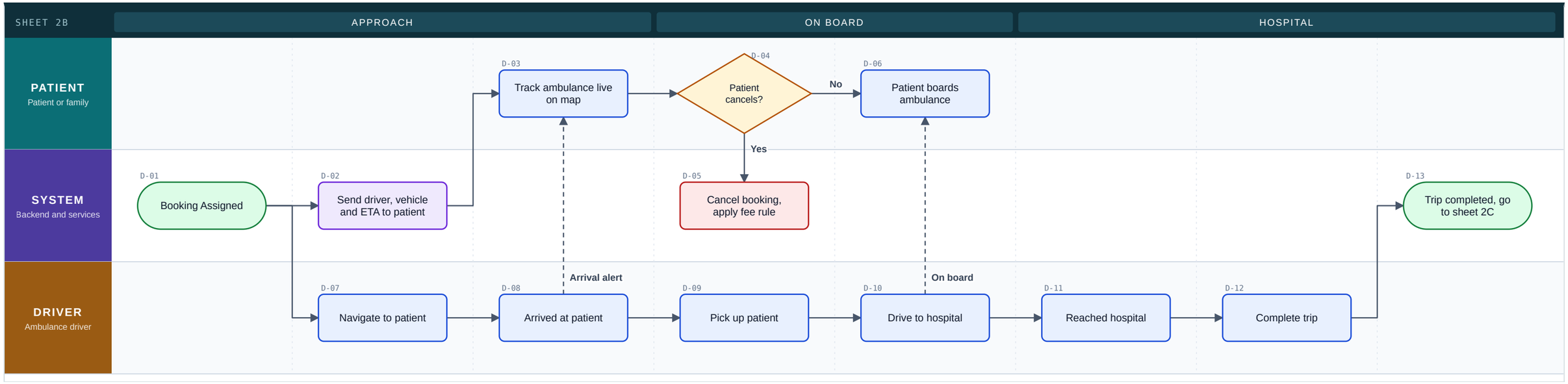
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
C-01	Patient	Choose type and filters	Type: General, ICU, CCU, NICU, patient transport. Filters: availability, nearest, fare, rating.
C-02	System	Search ambulances within radius	GET /ambulances/nearby with lat, lng and type. Starts at 5 km.
C-05	System	20 km reached: show helpline	Emergency helpline number shown instead of results.
C-06	Patient	Review results, ETA, fare, rating	Map and list, with distance, ETA, estimated fare, driver rating, equipment.
C-08	Patient	Pick hospital, set priority	Hospital is optional. Priority is Normal, Emergency or Critical.
C-09	System	Create booking, status Pending	POST /bookings. Fare estimate and booking id are created.
C-10	Driver	Driver gets alert, views details	Socket event plus push notification, with sound.
C-13	System	Retries left?	Maximum of 3 re-dispatch attempts (assumed).
C-14	System	Alert next nearest driver	Next candidate from the ranked list on sheet 5.
C-15	System	No driver free, patient told	Patient returns to the results list.

PATIENT · SYSTEM · DRIVER

BOOKING: THE TRIP

Driver status changes drive everything the patient sees. The patient can cancel until pickup. A fee applies only if the driver is already moving.



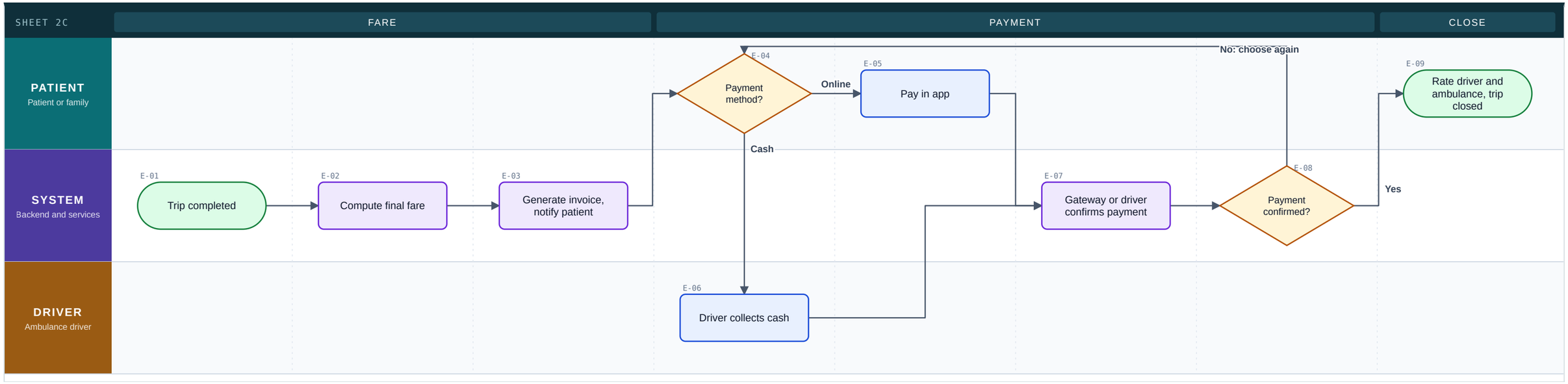
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
D-02	System	Send driver, vehicle and ETA to patient	Push notification and in-app card.
D-03	Patient	Track ambulance live on map	Marker moves with every 5 s update, see sheet 7.
D-05	System	Cancel booking, apply fee rule	Free if the driver has not started moving. Otherwise the fee set in Fare configuration applies.
D-07	Driver	Navigate to patient	Status EnRoute. Google Maps navigation.
D-08	Driver	Arrived at patient	Status Arrived. Patient is notified.
D-09	Driver	Pick up patient	Status PickedUp.
D-10	Driver	Drive to hospital	Status ToHospital. Route to the chosen destination.
D-11	Driver	Reached hospital	Status AtHospital.
D-12	Driver	Complete trip	Status Completed. Final GPS point is saved.

PATIENT · SYSTEM · DRIVER

BOOKING: FARE, PAYMENT AND RATING

The invoice is built from the finished trip. Cash is confirmed by the driver, card and mobile banking by the gateway. A failed payment sends the patient back to choose again.



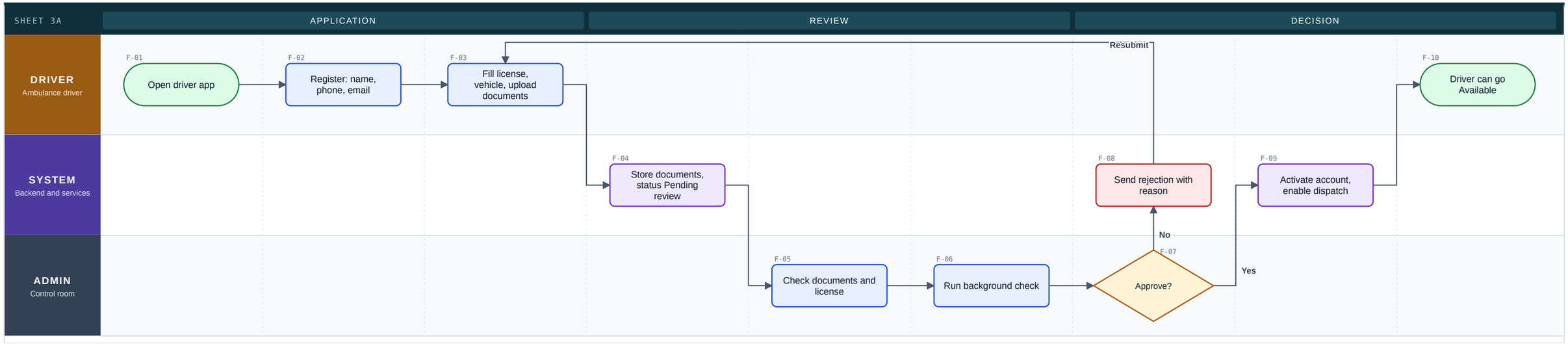
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
E-02	System	Compute final fare	Base fare + distance rate x km + waiting time.
E-03	System	Generate invoice, notify patient	Itemised invoice in the app.
E-05	Patient	Pay in app	Card or mobile banking through the payment gateway.
E-06	Driver	Driver collects cash	Driver confirms the amount received.
E-07	System	Gateway or driver confirms payment	Payment record saved: method, amount, status.
E-09	Patient	Rate driver and ambulance, trip closed	1 to 5 stars and an optional comment.

DRIVER · ADMIN

DRIVER ONBOARDING AND VERIFICATION

A driver cannot receive requests until an admin approves the documents. A rejection always carries a reason so the driver can fix the application and resubmit.



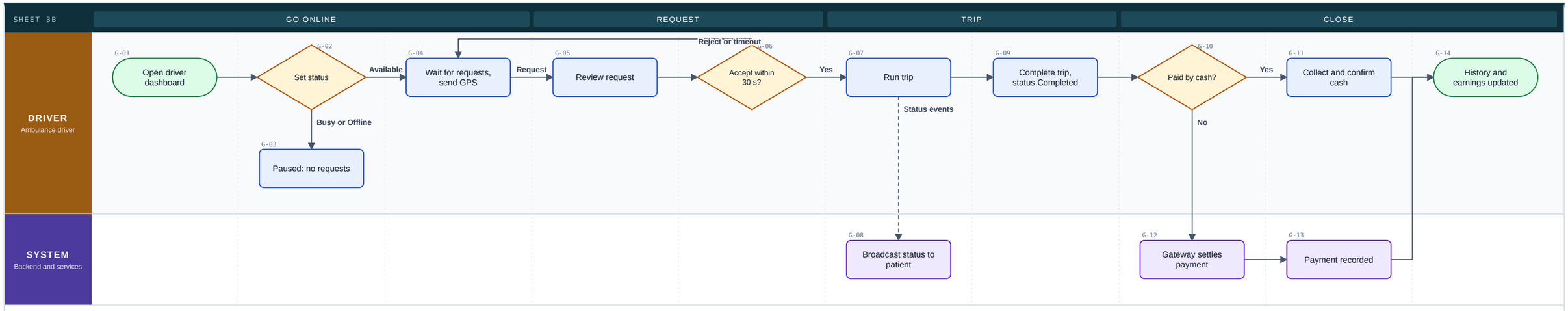
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
F-02	Driver	Register: name, phone, email	POST /drivers/register
F-03	Driver	Fill license, vehicle, upload documents	Driving license, ambulance details, registration papers.
F-04	System	Store documents, status Pending review	Files saved, review task created for admins.
F-05	Admin	Check documents and license	Valid license, vehicle registration, expiry dates.
F-06	Admin	Run background check	Identity and record check as required by local rules.
F-08	System	Send rejection with reason	Driver is notified and can resubmit.
F-09	System	Activate account, enable dispatch	Verification status set to Approved.

DRIVER

DRIVER DAILY OPERATION

Status controls whether a driver is dispatchable. Requests that are rejected or time out return the driver to waiting, with no penalty in this model.



STEP REFERENCE

ID	ACTOR	STEP	DETAIL
G-02	Driver	Set status	Available, Busy or Offline.
G-03	Driver	Paused: no requests	Busy means on another job. Offline turns off GPS and requests.
G-04	Driver	Wait for requests, send GPS	Location is sent every 5 s while Available.
G-05	Driver	Review request	Patient location, destination, ambulance type, priority.
G-07	Driver	Run trip	Navigate, pick up, deliver. See sheet 2B.
G-08	System	Broadcast status to patient	Each status change goes out over Socket.IO.
G-12	System	Gateway settles payment	Card and mobile banking are settled by the gateway.
G-13	System	Payment recorded	Payment status saved against the booking.

ADMIN

ADMIN CONTROL ROOM

Sign-in with optional two-factor, then a dashboard and seven modules. Each module lists the actions an admin can take.

Admin login – Two-factor code, if enabled – Dashboard: overview and statistics

USERS

- View all users
- Block or unblock
- See booking history

DRIVERS

- Verify documents
- Approve or reject, with reason
- Suspend a driver

AMBULANCES

- Add a vehicle
- Update type and equipment
- Cancel or retire a vehicle

HOSPITALS

- Add a hospital
- Update facilities and contact

BOOKINGS

- View all bookings
- Cancel a booking
- Resolve disputes

REPORTS

- Users, drivers, ambulances
- Bookings and completed trips
- Emergency requests, response time
- Revenue and fare charts

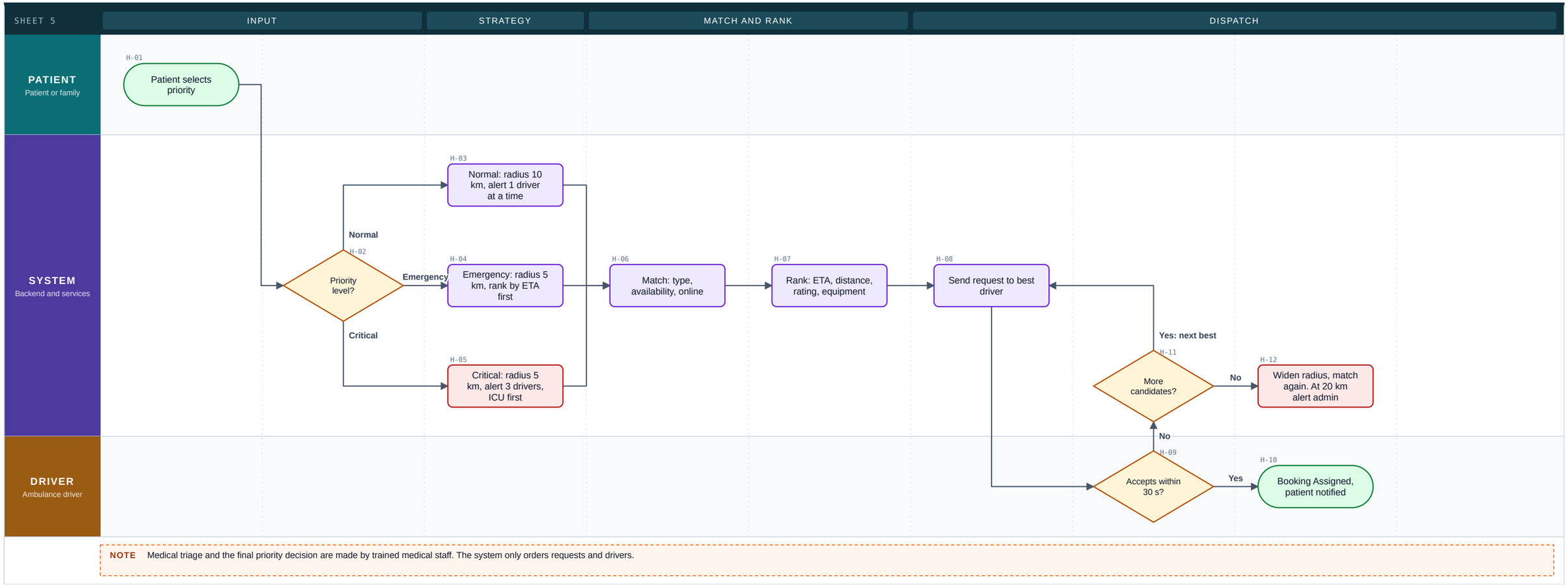
SYSTEM SETTINGS

- Emergency priority rules
- Fare configuration
- Notification templates
- API keys and map settings
- Security settings

SYSTEM

EMERGENCY PRIORITY AND DISPATCH

Priority changes how wide the search is and how many drivers are alerted together. Critical alerts three drivers at once and prefers ICU units.



NOTE Medical triage and the final priority decision are made by trained medical staff. The system only orders requests and drivers.

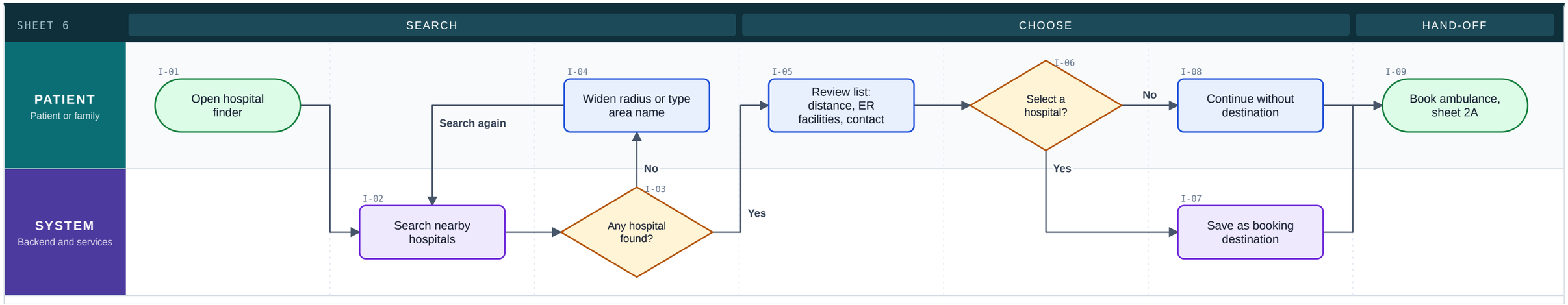
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
H-03	System	Normal: radius 10 km, alert 1 driver at a time	Standard transfer or non-urgent transport.
H-04	System	Emergency: radius 5 km, rank by ETA first	Fastest arrival wins over price and rating.
H-05	System	Critical: radius 5 km, alert 3 drivers, ICU first	First driver to accept gets the booking, the others are released.
H-06	System	Match: type, availability, online	Filter by ambulance type, Available status and a fresh GPS fix.
H-07	System	Rank: ETA, distance, rating, equipment	Sorted candidate list. Weights are set in Emergency priority rules.
H-08	System	Send request to best driver	Socket event plus push notification.
H-12	System	Widen radius, match again. At 20 km alert admin	Search radius grows one step per round.

PATIENT

HOSPITAL FINDER

Optional step. A chosen hospital becomes the trip destination. Skipping it lets the driver and patient decide on the way.



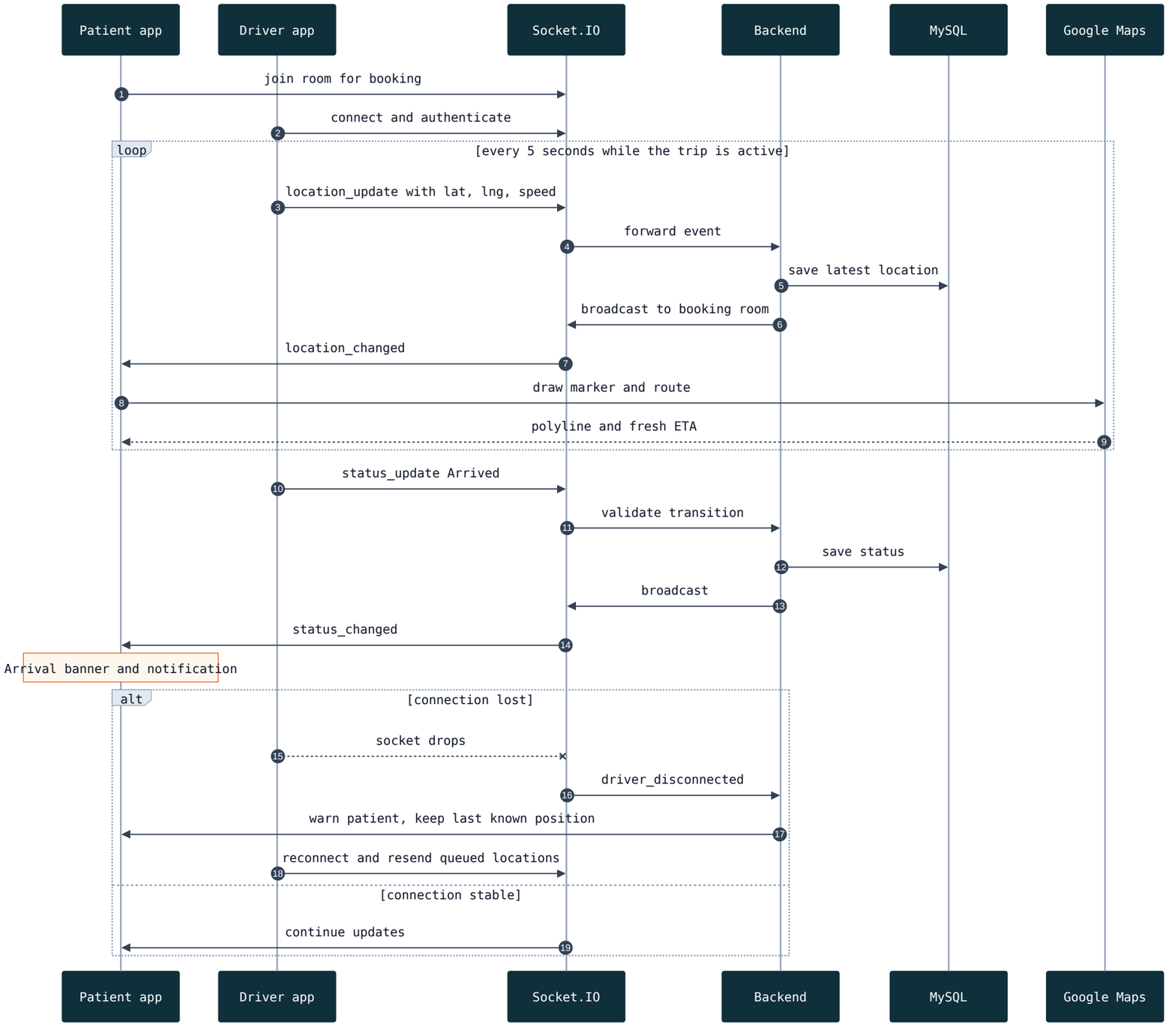
STEP REFERENCE

ID	ACTOR	STEP	DETAIL
I-02	System	Search nearby hospitals	Uses the confirmed location from sheet 1B.
I-05	Patient	Review list: distance, ER facilities, contact	Name, distance, emergency facilities, contact number, map pin.
I-07	System	Save as booking destination	Stored as the destination on the booking.

REAL-TIME

LIVE TRACKING, MESSAGE BY MESSAGE

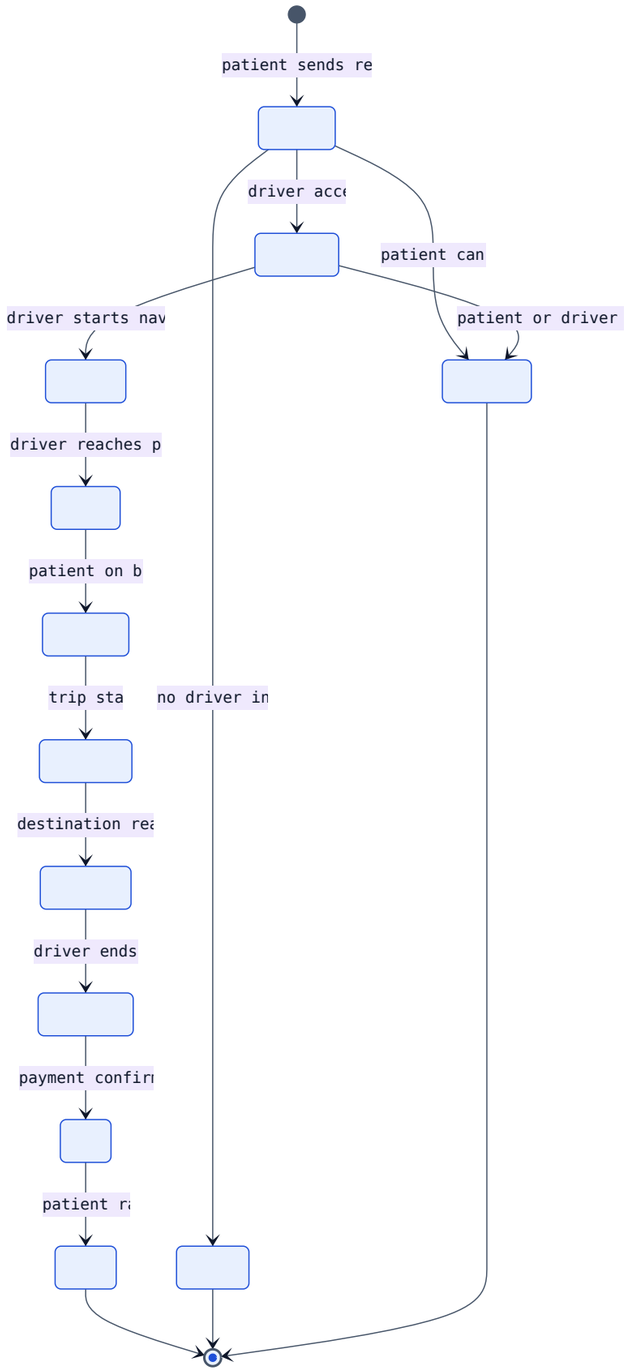
The driver app is the source of position. Socket.IO carries each update to the booking room, the backend stores it, and the patient map redraws. A dropped connection is handled explicitly.



BOOKING LIFECYCLE

TRIP STATUS MACHINE

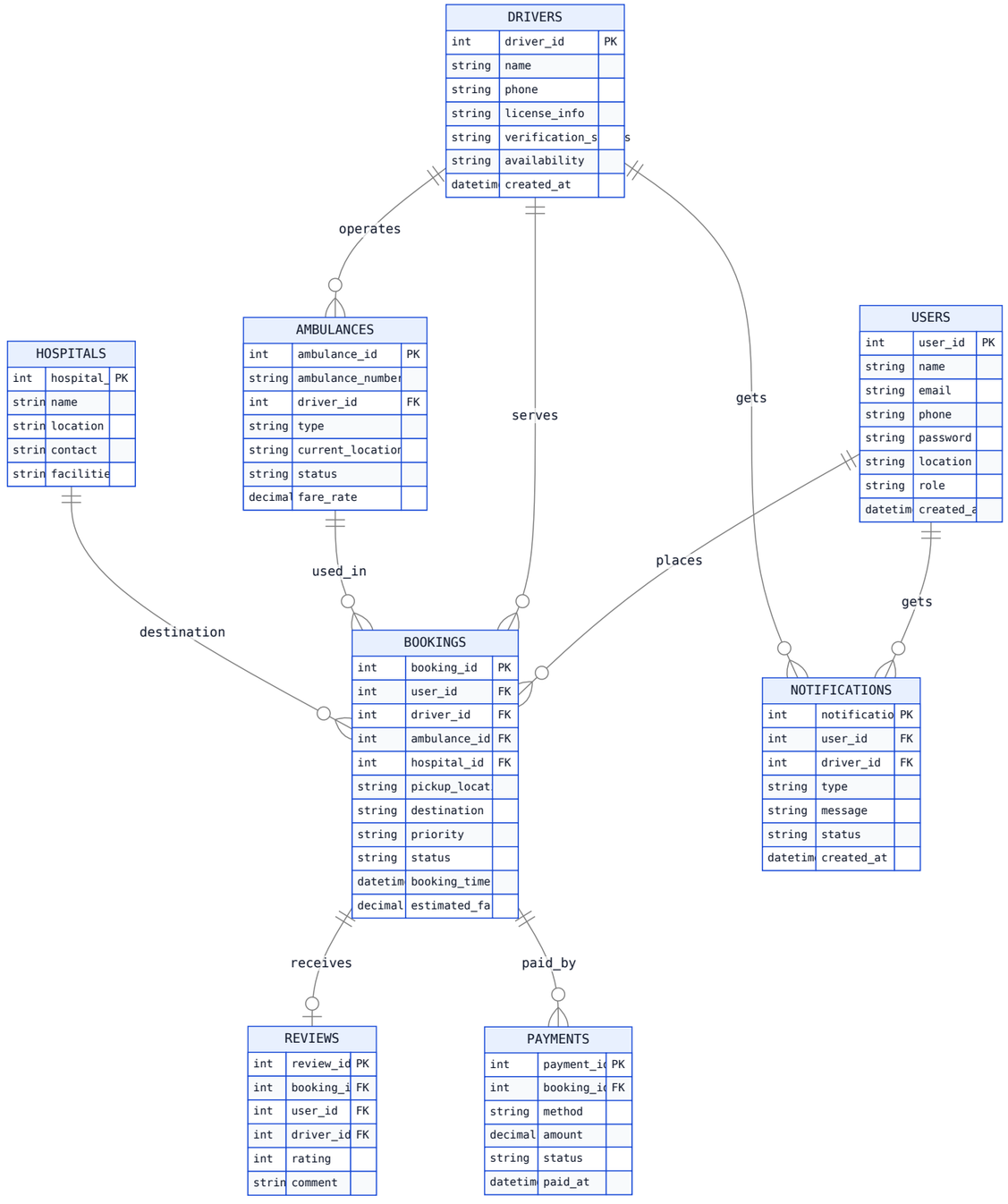
Every status a booking can hold and the event that moves it. Cancelled, Expired and Rated are final.



DATABASE

DATA MODEL

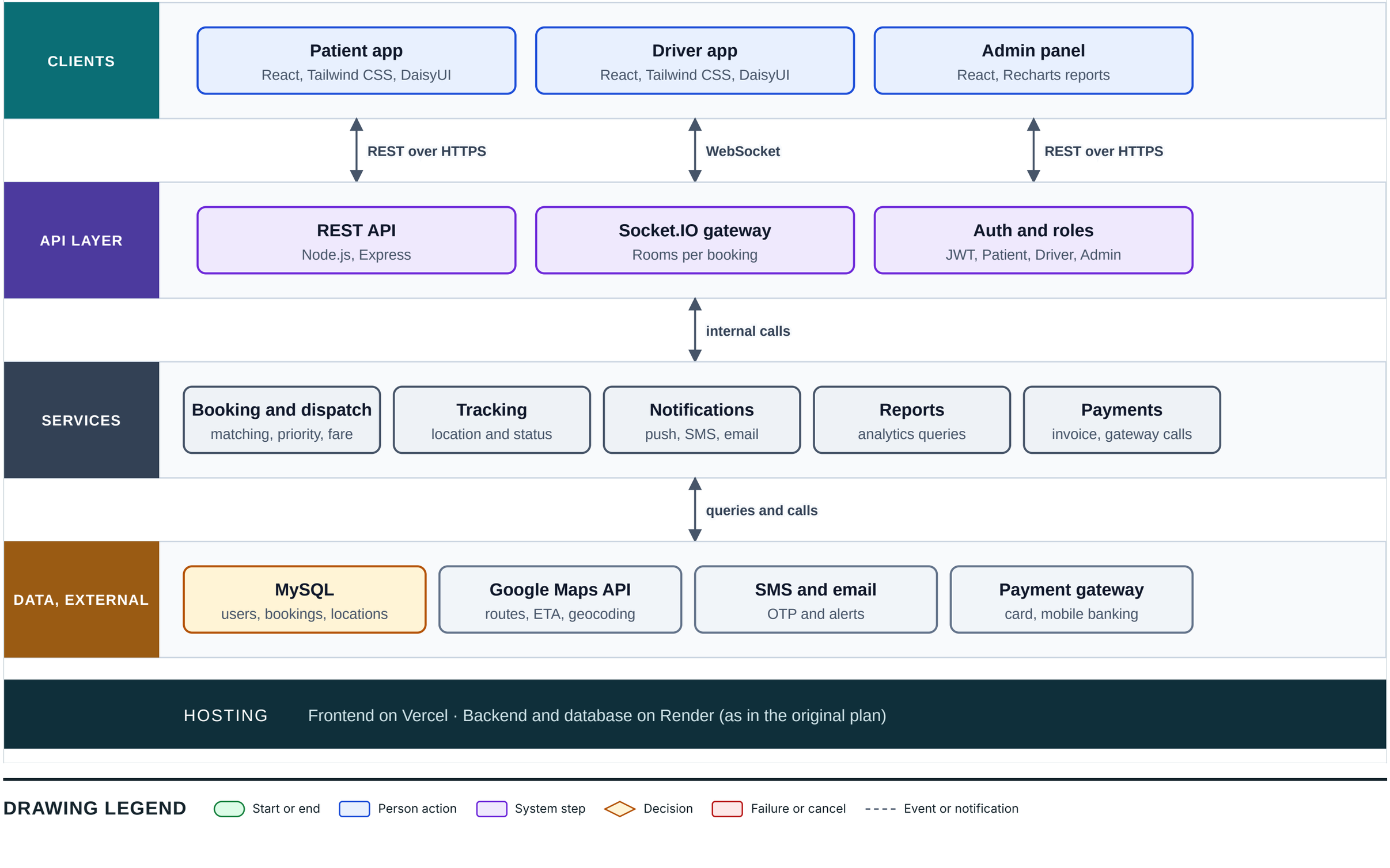
The six original tables, plus a payments table so invoices and gateway results have a home.



ARCHITECTURE

HOW THE PIECES CONNECT

Three apps share one backend. Real-time traffic goes through Socket.IO and everything else through REST. Maps, messaging and payments are external services.



DRAWING LEGEND Start or end · Person action · System step · Decision · Failure or cancel · ---- Event or notification